Т3.

$$\xi \sim p(x) = \begin{cases} \frac{\exp(-\frac{x}{\theta})}{\theta}, & x \geq 0 \\ 0, & x < 0 \end{cases}, \quad \theta > 0 \Rightarrow F(x) = \begin{cases} 1 - e^{-\frac{x}{\theta}}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$\underline{n = 3}$ $\quad \hat{\theta}_1 = \overline{X}, \quad \hat{\theta}_3 = X_{(2)}$

$$M\xi = \int_{-\infty}^{+\infty} x\,dF = \int_0^{+\infty} x \frac{e^{-\frac{x}{\theta}}}{\theta} dx = -\frac{x+\theta}{e^{\frac{x}{\theta}}}\Big|_0^{+\infty} = \theta$$

$$M\xi^2 = \int_{-\infty}^{+\infty} x^2 dF = \int_0^{+\infty} x^2 \frac{e^{-\frac{x}{\theta}}}{\theta} dx = \left\{ \begin{matrix} t = \frac{x}{\theta} \\ x = t\theta \end{matrix} \quad dx = \theta dt \right\} = \int_0^{+\infty} t^2 \theta^2 \frac{1}{\theta} e^{-t} \theta dt =$$

$$= \int_0^{+\infty} \theta^2 t^2 e^{-t} dt = -\theta^2 \int_0^{+\infty} t^2 d(e^{-t}) = -\theta^2 t^2 e^{-t}\Big|_0^{+\infty} + \theta^2 \int_0^{+\infty} 2t e^{-t} dt =$$

$$= -2\theta^2 \int_0^{+\infty} t\,d(e^{-t}) = -2\theta^2 t e^{-t}\Big|_0^{+\infty} + 2\theta^2 \int_0^{+\infty} e^{-t} dt = -2\theta^2 e^{-t}\Big|_0^{+\infty} =$$

$$= 2\theta^2 \cdot e^{-t}\Big|_{+\infty}^{0} = 2\theta^2$$

$$D\xi = M\xi^2 - (M\xi)^2 = 2\theta^2 - \theta^2 = \theta^2$$

a) $M\hat{\theta}_1 = M\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) = \frac{1}{n}\sum_{i=1}^{n} M\xi = \frac{1}{n} \cdot n M\xi = \theta \Rightarrow$

$\Rightarrow \hat{\theta}_1 = \overline{X}$ — несмещённая

$$M\hat{\theta}_3 = \int_{-\infty}^{+\infty} x \cdot \frac{3!}{(2-1)!(3-2)!} F(x) \cdot (1 - F(x)) p(x) dx =$$

$$= 6\int_0^{+\infty} x\left(1 - e^{-\frac{x}{\theta}}\right) e^{-\frac{x}{\theta}} \frac{1}{\theta} e^{-\frac{x}{\theta}} dx = \frac{6}{\theta}\int_0^{+\infty}\left(x e^{-\frac{2x}{\theta}} - x e^{-\frac{3x}{\theta}}\right) dx =$$

$$= \frac{6}{\theta}\left(-\frac{\theta}{2} x e^{-\frac{2x}{\theta}}\Big|_0^{+\infty} + \frac{\theta}{2}\int_0^{+\infty} e^{-\frac{2x}{\theta}} dx + \frac{\theta}{3} x e^{-\frac{3x}{\theta}}\Big|_0^{+\infty} - \frac{\theta}{3}\int_0^{+\infty} e^{-\frac{3x}{\theta}} dx\right) =$$

$$= \frac{6}{\theta}\left(-\frac{\theta^2}{4} e^{-\frac{2x}{\theta}}\Big|_0^{+\infty} + \frac{\theta^2}{9} \cdot e^{-\frac{3x}{\theta}}\Big|_0^{+\infty}\right) = \frac{6}{\theta}\left(\frac{\theta^2}{4} - \frac{\theta^2}{9}\right) = \frac{5}{6}\theta \Rightarrow$$

$\Rightarrow \hat{\theta}_3$ — смещённая $\Rightarrow \hat{\theta}_3' = \frac{6}{5} X_{(2)}$

b) $D\hat{\theta}_1 = \frac{1}{n} D\xi = \frac{1}{n}\theta^2 = \frac{\theta^2}{3}$

$$M\hat{\theta}_3^2 = \int_{-\infty}^{+\infty} x^2 \frac{3!}{(2-1)!(3-2)!} F(x)(1 - F(x)) p(x) dx = \frac{6}{\theta}\int_0^{+\infty}\left(x^2 e^{-\frac{2x}{\theta}} - x^2 e^{-\frac{3x}{\theta}}\right) dx =$$

$$= \frac{6}{\theta}\left(-\frac{\theta}{2} x^2 e^{-\frac{2x}{\theta}}\Big|_0^{+\infty} + \frac{\theta}{2}\int_0^{+\infty} e^{-\frac{2x}{\theta}} 2x\,dx + \frac{\theta}{3} x^2 e^{-\frac{3x}{\theta}}\Big|_0^{+\infty} - \frac{\theta}{3}\int_0^{+\infty} e^{-\frac{3x}{\theta}} 2x\,dx =\right.$$

$$= \frac{6}{\theta}\left(\theta\int_0^{+\infty} x e^{-\frac{2x}{\theta}}dx - \frac{2\theta}{3}\int_0^{+\infty} x e^{-\frac{3x}{\theta}}dx\right) =$$

$$= \frac{6}{\theta}\left(\theta\cdot\frac{\theta^2}{4} - \frac{2\theta}{3}\cdot\frac{\theta^2}{9}\right) = 6\theta^2\cdot\frac{19}{108} = \frac{19}{18}\theta^2$$

$$D\tilde{\theta}_3 = M\tilde{\theta}_3^2 - (M\tilde{\theta}_3)^2 = \frac{19}{18}\theta^2 - \frac{25}{36}\theta^2 = \frac{13}{36}\theta^2$$

$$D\tilde{\theta}_3' = \left(\frac{6}{5}\right)^2 D\tilde{\theta}_3 = \frac{36}{25}\cdot\frac{13}{36}\theta^2 = \frac{13}{25}\theta^2$$

$D\tilde{\theta}_1 = \frac{\theta^2}{3} < \frac{13}{25}\theta^2 = D\tilde{\theta}_3' \Rightarrow \tilde{\theta}_1$ - более эффективна

c) Сначала проверим модель на регулярность:

$p(x,\theta) = \frac{1}{\theta}e^{-\frac{x}{\theta}}\{[0;+\infty)\}, \theta>0$ - непр. дифф.

$\ln p = -\ln\theta - \frac{x}{\theta}$ $\quad \frac{\partial \ln p}{\partial\theta} = -\frac{1}{\theta} + \frac{x}{\theta^2}$

$$\int_0^{+\infty}\frac{\partial p(x,\theta)}{\partial\theta}dx = \int_0^{+\infty}\left(-\frac{1}{\theta^2}e^{-\frac{x}{\theta}} + \frac{1}{\theta}\cdot x\cdot\frac{1}{\theta^2}e^{-\frac{x}{\theta}}\right)dx =$$

$$= \frac{1}{\theta^2}\int_0^{+\infty}\left(\frac{x}{\theta}-1\right)e^{-\frac{x}{\theta}}dx = \frac{1}{\theta^2}\left(\int_0^{+\infty}\frac{x}{\theta}e^{-\frac{x}{\theta}}dx - \int_0^{+\infty}e^{-\frac{x}{\theta}}dx\right) =$$

$$= \frac{1}{\theta^2}\left(\theta + \theta e^{-\frac{x}{\theta}}\Big|_0^{+\infty}\right) = \frac{1}{\theta^2}(\theta-\theta) = 0$$

$$I(\theta) = \int_0^{+\infty}\left(\frac{\partial \ln p}{\partial\theta}\right)^2 p(x,\theta)dx = \int_0^{+\infty}\left(-\frac{1}{\theta}+\frac{x}{\theta^2}\right)^2\cdot\frac{1}{\theta}e^{-\frac{x}{\theta}}dx =$$

$$= \int_0^{+\infty}\left(\frac{1}{\theta^2} - 2\frac{x}{\theta^3} + \frac{x^2}{\theta^4}\right)\frac{1}{\theta}e^{-\frac{x}{\theta}}dx = \frac{1}{\theta^3}\int_0^{+\infty}e^{-\frac{x}{\theta}}dx - \frac{2}{\theta^3}\int_0^{+\infty}\frac{x}{\theta}e^{-\frac{x}{\theta}}dx +$$

$$+ \frac{1}{\theta^4}\int_0^{+\infty}\frac{x^2}{\theta}e^{-\frac{x}{\theta}}dx = \frac{1}{\theta^3}\cdot\theta - \frac{2}{\theta^3}\cdot\theta + \frac{1}{\theta^4}\cdot 2\theta^2 = \frac{1}{\theta^2} - \frac{2}{\theta^2} + \frac{2}{\theta^2} = \frac{1}{\theta^2}$$

$I(\theta) > 0$ на Θ и непр. $\Rightarrow$ модель регулярна

$\tilde{\theta}_1 = \overline{x}$ - несмещённая и $D\tilde{\theta}_1 = \frac{\theta^2}{3}$ - огран. на $\forall$ компакте из $\Theta \Rightarrow \tilde{\theta}_1$ - регулярная оценка

$\tilde{\theta}_3' = \frac{6}{5}x_{(2)}$ - несмещённая и $D\tilde{\theta}_3' = \frac{13}{25}\tilde{\theta}_3'$ - огран. на $\forall$ компакте из $\Theta \Rightarrow \tilde{\theta}_3'$ - регулярная оценка

Тогда и модель, и оценки являются регулярными => можно использовать нер-во Крамера-Рао.

Для $\tilde{\theta}_1$: $D\tilde{\theta}_1 = \frac{\theta^2}{3} = \frac{1}{nI(\theta)} = \frac{1}{\frac{1}{n}\cdot\frac{1}{\theta^2}}\Big|_{n=3} = \frac{\theta^2}{3}$ =>

=> $\tilde{\theta}_1$ – эффективна

Для $\tilde{\theta}_3'$: $D\tilde{\theta}_3' = \frac{13}{25}\theta^2 > \frac{1}{nI(\theta)}\Big|_{n=3} = \frac{\theta^2}{3}$ => из нер-ва ничего не следует.